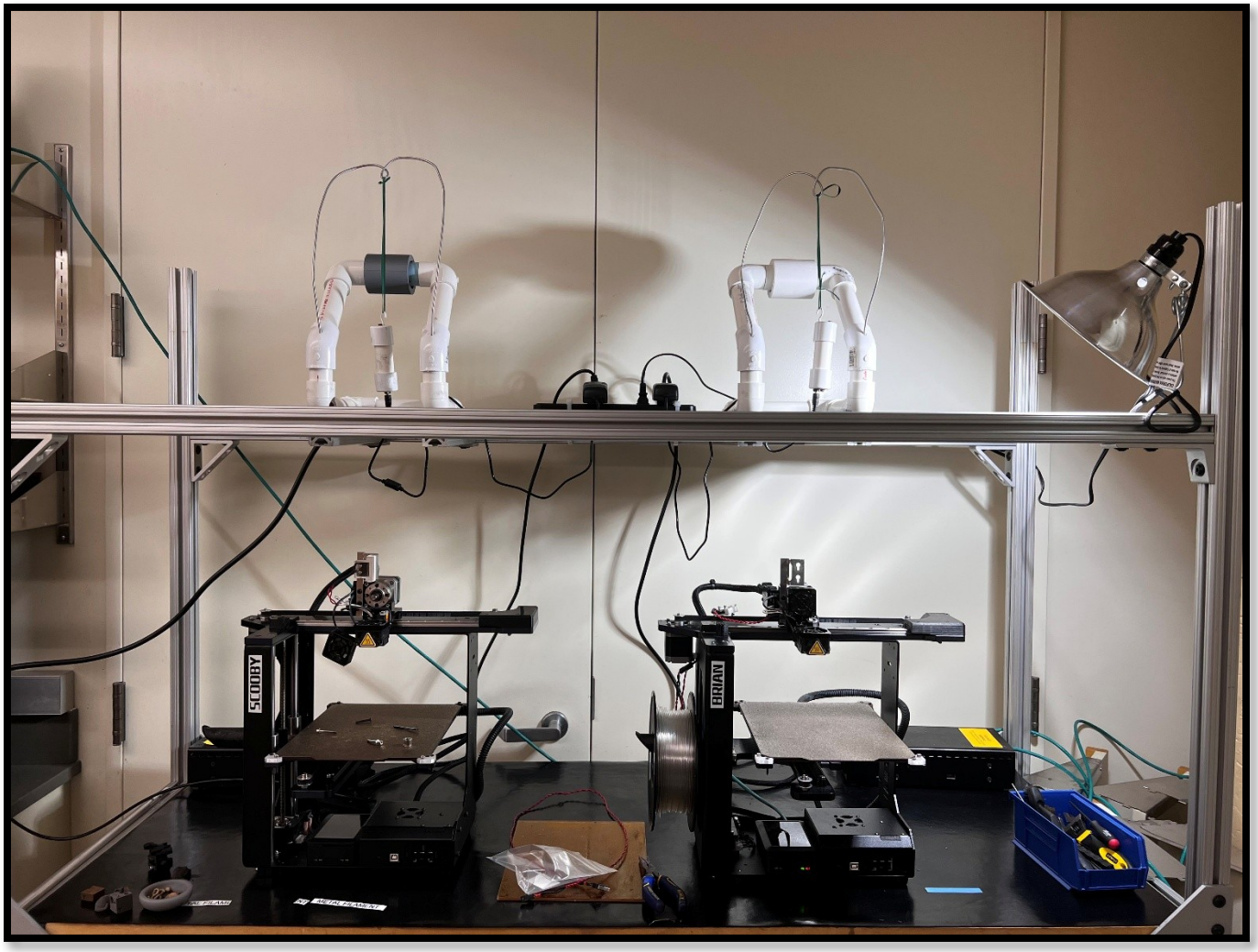
METAL 3D-PRINTING

OPERATIONS GUIDE



Boston University College of Engineering
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Introduction

For metal3d printing at EPIC, we use a method called Bound Metal Deposition (BMD), which is sometimes referred to as Metal Fused Filament Deposition. For the purposes of this document, I will refer to the process as BMD.

Bound Metal Deposition is often the most accessible form of metal 3d printing as it does not require any special printing technology, however, post-processing does require a high temp furnace.

BMD is the process of using a metal powder, and a binder (typically PLA or PETG) to form a filament where the metal powder is “Bound” by the binder. Because the filament is mostly PLA or PETG, it can be printed using similar print settings to the settings for those materials. This is what makes BMD so accessible, once you have the filament, you can print it on just about any 3d printer with only a few caveats.

Caveat 1: Brittleness

While the filaments can be printed with similar settings to their fully plastic counterparts, the metal powder does change the mechanical properties of the filament. This is most apparent in the filament’s brittleness. The BMD filament is often so brittle in fact, that it must be softened before it leaves the spool as the unbending of the filament from the spool is enough to break the filament. The filament also tends to break when the nozzle rapidly moves side to side.

Solution 1: Vertical Filament Feeding

The first problem to fix, is the way in which the filament is fed into the printer. Since most 3d printers hold the spool on the side of the printer and feed the filament over the top to the nozzle, a custom spool holder had to be made. Luckily, the manufacturer that makes our BMD filament, The Virtual Foundry, makes a custom spool holder to feed filament vertically, straight downward to the nozzle.

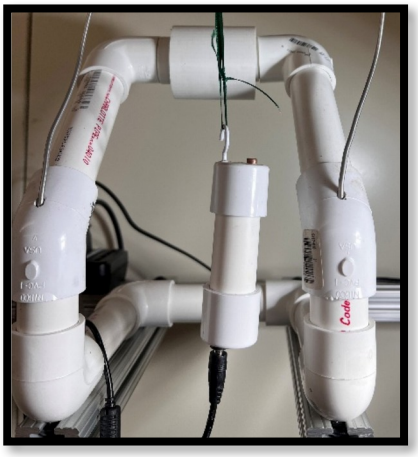

Figure 1

Figure 2

Solution 2: FilaWarmer

The manufacturer that supplies our BMD filament, The Virtual Foundry, makes a complimentary product to their filament, the FilaWarmer. The FilaWarmer heats the filament coming off the spool, softening it so that the straightening of the filament doesn't cause the filament to break. Soon after the filament leaves the warmer, it becomes brittle again, so it was crucial not only to use the FilaWarmer, but also to have it mounted directly above the nozzle, to allow for minimal bending between the warmer and the nozzle

Caveat 2: Clogging

This is the largest problem we ran into, one that doesn't have a clear solution. When the filament enters the hot end, it heats and melts. Typically, with normal PLA, a small amount of the plastic can off gas into the atmosphere, the amount is usually minuscule, and poses no harm. However, since metal powder is embedded in the PLA, and the metal powder has a high thermal conductivity, the filaments seems to heat more unevenly, causing much more off gassing. While the burned filament has some toxicity, in these amounts, health concerns remain minimal. Where this does have a large effect, is in nozzle clogging.

When a substantial amount of the PLA in the mixed filament burns off, it leaves behind the metal powder, which sticks together, and forms a hard, solid layer. This layer cannot be melted off, so when this occurs, the nozzle assembly must be taken apart, and the fused powder must be drilled out, which has potential to damage the nozzle (However if careful damage is avoidable).

Solution 1: Custom G-Code

To attempt to mitigate nozzle clogging, we have implemented custom G-Code, where the printer first completes all initialization steps (Bed Leveling, Heating, etc.), it pauses the print. While the print is paused the user feeds the filament through the nozzle, and purges the nozzle of any leftover filament. Once this is done, the user resumes the print. This allows for the filament to be in the nozzle, without extruding for the least amount of time.

Solution 2: Post Printing

After printing is finished, the metal filament should be immediately removed from the nozzle, and the nozzle should be purged with PLA. This again ensures the metal filament is in the nozzle without extrusion for the least amount of time.

Note: The Metal Filament should never be in the nozzle while not printing. If the metal filament cools inside the nozzle, it will almost certainly clog.

Setup

POWER UP & LOGIN

Power Up & Login

Figure 3: Power Supply

1. Turn on the Power Supply behind the printer. You should hear a couple of beeps as soon as it turns on, if you don't, ensure that the power strip is powered on, and that all connections are tight.
2. On the computer next to the printers, open Google Chrome (ensure that you are logged into the epic3d@bu.edu account)
3. On the bookmark bar, click on the folder labeled "Maker Gears" and select the printer you are using (either Scooby or Brian)
4. If they haven't been used recently, the website may prompt you to login.
    a. Username: "printer"
    b. Password: "epic"

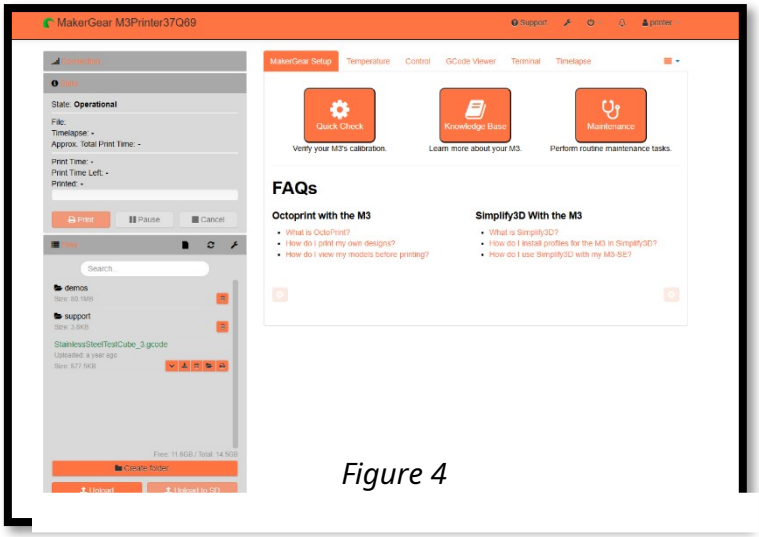

Figure 4

5. You should now be at the page shown in Figure 3

SETUP & CALIBRATION

1. Click the “Maintenance” button
2. Open the “Starting Height Adjustments” drop down menu
3. Click “Tool 0 (left) Starting Height Cold – Probe”
4. Click “Prepare”
5. This should trigger the printer to move into place for the z axis calibration. Once finished, their should be a noticeable space (≈ 0.25” – 0.5”) between the bed and the nozzle. If the gap is very large, or very small, stop here and get assistance, as this signals a disconnected probe.

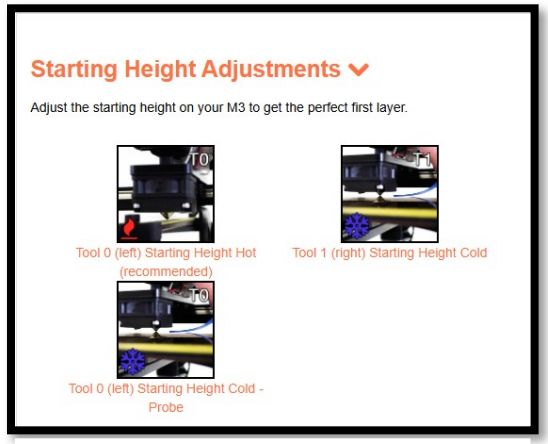

Figure 5

6. Once the printer stops, scroll down and you should see a section labeled “Bed Height Buttons”. Once you are here, follow these steps:
   a. Place the blue calibration strip directly underneath the nozzle
   b. Press the “1mm Up” button, then check to see if the paper is stuck under the nozzle. If not, repeat this until the paper is stuck under the nozzle. Once this happens, press the “1 mm Down” button once

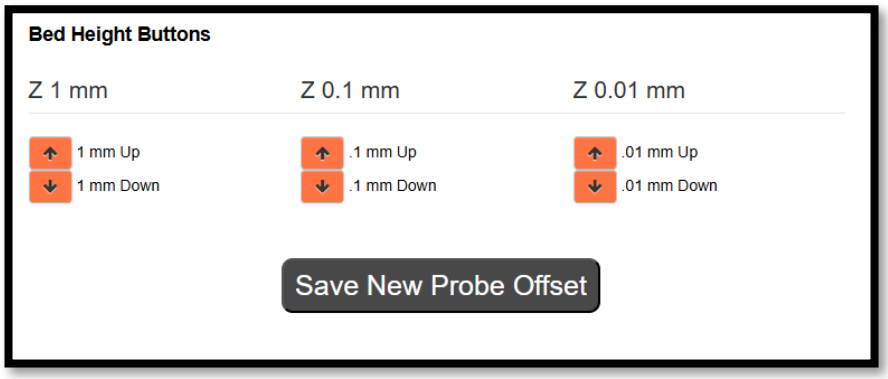

Figure 6

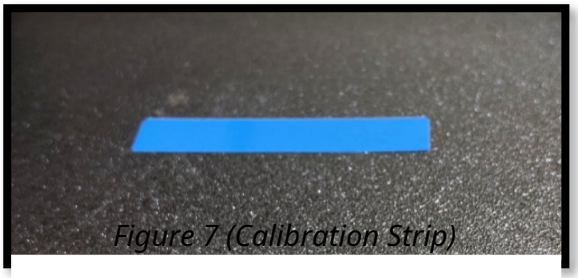

Figure 7 (Calibration Strip)

c. Next, press the “0.1 mm Up” button, until the paper is stuck under the nozzle. Once this happens, press “0.1 mm Down”

d. Press the “0.01 mm Up” button once.

e. Click “Save New Probe Offset”

7. Navigate to the “Temperature” section at the top of the page. Click on the drop down menu, next to the check mark, in the row titled “Tool”, select “Set PLA (215°C)”. Wait for the nozzle to reach the desired Temperature.

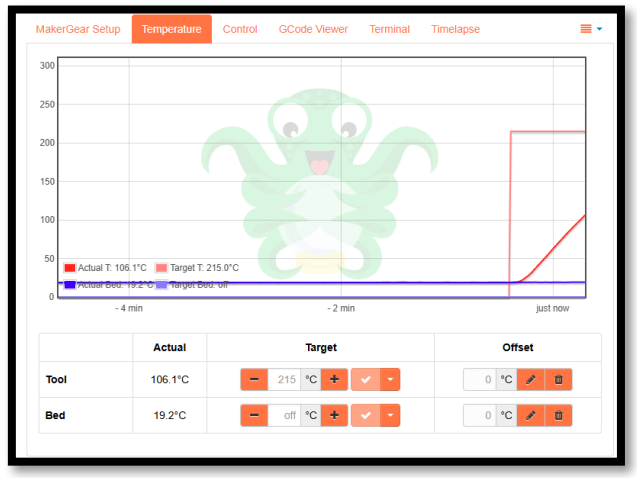

Figure 6

8. Navigate to the “Control” section at the top of the page. Click the up arrow in the Z direction several times, to lower the bed to a reasonable height.
9. Click the right arrow in the X/Y controls to move the nozzle to the center of the bed
10. Use the PLA filament held on the side of one of the printers. Remove the runout sensor from on top of the nozzle assembly. Place the PLA filament into the hole at the top of the nozzle and feed it in. (the filament tends to get caught along the way, so this takes some wiggling). Once the filament is in, press the extrude button several times, to form a small pool of filament on the bed.
11. Inspect the filament that has been extruded. If it is perfectly clear, and it extruded smoothly, your nozzle is good. If there are metallic dots coming out in the clear filament, continue extruding. If the metal dots don’t go away, the nozzle is likely starting to clog.
12. If the nozzle is in good condition, press the “Retract” button to retract the PLA completely out of the nozzle assembly.

Printing

PRINTING

1. Open the Prusa Slicer on the computer

2. In the top left-hand corner, click “file”, then click “Open Project”
    a. If using Bronze Filament. Copy and paste the following file path: E:\Public\3DP-Jira\3DPLab_Makergear\BronzeFilamet_Template.3mf
    b. If using Stainless Steel Filament, copy and paste the following file path: E:\Public\3DP-Jira\3DPLab_Makergear\StainlessSteelFilamet_Template.3mf
3. Click “Open”
4. Upload your STL/3MF file, and setup your print
    a. Do not change the layer height, speed, filament settings, and custom G-Code
5. Once you have sliced, and downloaded your G-Code file, click Save As (DO NOT CLICK SAVE), and save your file.
6. You can now exit Prusa Slicer, and return to the Printer Website.
7. Click the “Upload” button, and select your G-Code file.
8. Click “Print”
9. The printer will now go through its initialization. Allow the printer to come to a complete stop for several seconds. The printer is now in pause mode, and will not resume until the user clicks resume.
10. Guide the metal filament through the Filament warmer, through the runout sensor, and into the nozzle assembly.
11. Click the extrude button repeatedly, until the filament is uniformly coming out of the nozzle, and there is no clear filament coming out of the nozzle.
12. Attach the runout sensor to the snap in place fixture above the nozzle assembly. The Printer should now match Figure 8.
13. Click “Resume”

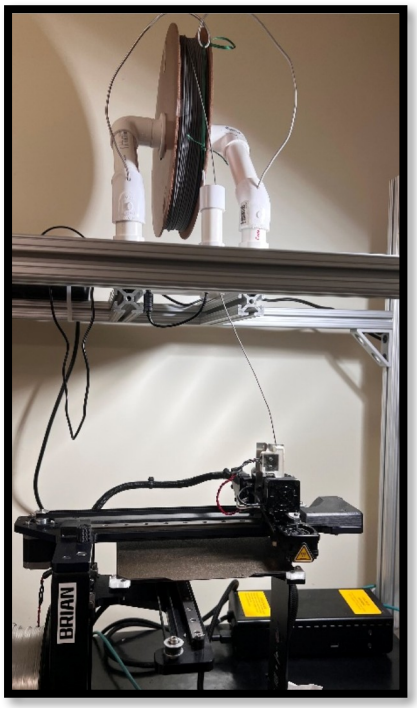

Figure 7

Notes:

- During the printing process, filament may build up around the edges of the nozzle, this typically does not cause problems, however if enough builds up it can cause errors. If this happens, pause the print, move the nozzle out of the way, and wipe it down. Then click “resume”

- When using bronze filament, sometimes the filament gets stuck, and the drive gears no longer push the filament. If this happens, it may be too late, however if you catch it early enough you can manually push the filament into the nozzle assembly to continue extruding, once you do this it should go back to automatically extruding

AFTER PRINTING

1. Immediately go into the temperature section, and set the Tool temperature to the PLA setting (after printing the nozzle automatically cools)
2. Click retract several times until the filament exits the nozzle assembly.
3. Perform a purge using the clear PLA, following the same steps from the Setup section.

Shutdown Procedure:

Note: Never directly turn the Power Supply off before following these steps

1. In the printer website, in the top right-hand corner, click the power button
2. Click “Shutdown system”. A popup will appear asking if you are sure. Click “Proceed”
3. You should hear two beeps, followed by a short melody. Once that melody plays, it is safe to turn the Power Supply off.
4. Remove the metal filament from the spool holder and place it in the plastic box below the printers to keep moister out.
5. Turn off the Power strip.

Post Processing

Figure 8: Furnace located in EPC B23

Nomenclature: I will use the typical nomenclature for Metal 3D Printed Parts for the rest of this documentation. The nomenclature follows:

- Green Part: your 3d printed part directly after printing, before debinding
- Brown Part: your part after debinding, but before sintering
- Sintered Part/Final Part: your part post sintering, final mechanical properties.

DEBINDING

<u>Bronze</u>

1. Locate the following items:
    a. Stainless Steel Crucible
    b. Al_2O_3 Powder
    c. Your Green Part
2. Bury your green part:
    a. Fill the crucible about ¼ of the

<u>Stainless Steel</u>

1. Locate the following items:
    a. Graphite Crucible
    b. Steel Blend Refractory Powder
    c. Your Green Part
2. Bury your green part:
    a. Fill the crucible about ¼ of the

way to the top with the Al_2O_3 Powder.

b. Place your green part on top of the Al_2O_3 Powder inside the crucible

c. Cover your part with the Al_2O_3 Powder, leaving about ½" between the top of the crucible and the powder

d. Softly tap the crucible several times to ensure the powder fills all cracks and crevices of the print

3. Place the Crucible Into the Furnace
4. Programming the Furnace:

a. Turn on the furnace (switch on side of control panel)

b. Press "Start"

c. A red light next to "Program" should light up. If it doesn't, Press "Start" again until it does.

d.

e. The display should now say: "Pr" followed by a number. Press the up arrow until it displays "Pr01"

f. Press "Start", "Ra1" and a number should begin blinking

g. Use the up and down arrows to way to the top with the Steel Blend Refractory Powder

b. Place your green part on top of the Steel Blend Refractory Powder inside the crucible

i. Ensure that at least ½" of powder is between your part and the crucible on all sides

c. Cover your part with the Steel Blend Refractory Powder.

d. Softly tap the crucible several times to ensure the powder fills all cracks and crevices of the print

3. Place the Crucible Into the Furnace
4. Programming the Furnace:

a. Turn on the furnace (switch on side of control panel)

b. Press "Start"

c. A red light next to "Program" should light up. If it doesn't, Press "Start" again until it does.

d. The display should now say: "Pr" followed by a number. Press the up arrow until it displays "Pr02"

e. Press "Start", "Ra1" and a number should begin blinking

change the numerical value to 100.

h. Press “Start”, “°F” and a number should begin blinking

i. Change the value to 900.

j. Press “Start”, “Hld1” and a number should begin blinking

k. Change the value to 2

l. Press “Start”, “Ra2” and a number should begin blinking.

m. Change the value to 0000

n. Press “Start”, “Start” should appear on the display again.

o. Ensure that the lid is on the furnace with no gaps, and the Fume hood is turned on and lowered

p. Press “Start” to begin Debinding process.

5. After Debinding is complete, carefully dump your part out, dispose of left over powder

f. Use the up and down arrows to change the numerical value to 200.

g. Press “Start”, “°F” and a number should begin blinking

h. Change the value to 400.

i. Press “Start”, “Hld1” and a number should begin blinking

j. Change the value to 2

k. Press “Start”, “Ra2” and a number should begin blinking.

l. Change the value to 200

m. Press “Start”, “°F” and a number should begin blinking

n. Change the value to 800.

o. Press “Start”, “Hld2” and a number should begin blinking

p. Change the value to 2

q. Press “Start”, “Ra3” and a number should begin blinking.

r. Change the value to 0000

s. Press “Start”, “Start” should appear on the display again.

t. Ensure that the lid is on the furnace with no gaps, and the Fume hood is turned on and

lowered

u. Press “Start” to begin Debinding process.

5. After Debinding is complete, do not dump the crucible, carefully place the full crucible, with the refractory powder, on the table to prepare for sintering.

SINTERING

Bronze

1. Locate the following items:
    a. Stainless Steel Crucible
    b. Talc Powder
    c. Sintering Carbon
    d. Kiln Paper/Ceramic Slab
    e. Your Brown Part
2. Bury your brown part:
    a. Fill the crucible about ¼ of the way to the top with the Talc Powder
    b. Place your green part on top of the Talc Powder inside the crucible
    c. Cover your part with the Talc Powder, leaving about ½” between the top of the crucible

Stainless Steel

1. Locate the following items:
    a. Sintering Carbon
2. Fill the remaining space of the crucible with sintering powder
3. Place the Crucible Into the Furnace
4. Programming the Furnace:
    a. Turn on the furnace (switch on side of control panel)
    b. Press “Start”
    c. A red light next to “Program” should light up. If it doesn’t, Press “Start” again until it does.
    d. The display should now say: “Pr” followed by a number. Press the up arrow until it displays “Pr02”
    e. Press “Start”, “Ra1” and a

and the powder

d. Add about ½" of sintering carbon on top of the Talc Powder

e. Softly tap the crucible several times to ensure the powder fills all cracks and crevices of the print

3. Place the crucible into the furnace, and place the kiln paper/ceramic slab on top of the crucible, so that the top is completely covered.

4. Program the furnace:

a. Turn on the furnace (switch on side of control panel)

b. Press "Start"

c. A red light next to "Program" should light up. If it doesn't, Press "Start" again until it does.

d. The display should now say: "Pr" followed by a number. Press the up arrow until it displays "Pr03"

e. Press "Start", "Ra1" and a number should begin blinking

f. Use the up and down arrows to change the numerical value to 200.

g. Press "Start", "°F" and a number number should begin blinking

f. Use the up and down arrows to change the numerical value to its maximum, it should say "FULL"

g. Press "Start", "°F" and a number should begin blinking

h. Change the value to 1100.

i. Press "Start", "Hld1" and a number should begin blinking

j. Change the value to 2

k. Press "Start", "Ra2" and a number should begin blinking.

l. Change the value to 575

m. Press "Start", "°F" and a number should begin blinking

n. Change the value to 2250.

o. Press "Start", "Hld2" and a number should begin blinking

p. Change the value to 4

q. Press "Start", "Ra3" and a number should begin blinking.

r. Change the value to 190

s. Press "Start", "°F" and a number should begin blinking

t. Change the value to 1100.

u. Press "Start", "Hld3" and a

should begin blinking

h. Change the value to 1625.

i. Press “Start”, “Hld1” and a number should begin blinking

j. Change the value to 2

k. Press “Start”, “Ra2” and a number should begin blinking.

l. Change the value to 0000

m. Press “Start”, “Start” should appear on the display again.

n. Ensure that the lid is on the furnace with no gaps, and the Fume hood is turned on and lowered

o. Press “Start” to begin Sintering process.

5. Once finished, dump your part out of the crucible, your part is now fully Sintered!

number should begin blinking

v. Change the value to 0

w. Press “Start”, “Ra4” and a number should begin blinking.

x. Change the value to 0000

y. Press “Start”, “Start” should appear on the display again.

z. Ensure that the lid is on the furnace with no gaps, and the Fume hood is turned on and lowered

aa. Press “Start” to begin Sintering process.

5. Once finished, dump your part out of the crucible, your part is now fully Sintered!

Appendices

FURNACE PROGRAMS

	Bronze Debinding	Bronze Sintering	Stainless Steel Debinding	Stainless Steel Sintering
Operation	Value	Value	Value	Value
Ra1	100	200	200	FULL
°F	900	1625	400	1100
Hld1	2	2	2	2
Ra2	0	0	200	575
°F	N/A	N/A	800	2250
Hld2	N/A	N/A	2	4
Ra3	N/A	N/A	0	190
°F	N/A	N/A	N/A	1100
Hld3	N/A	N/A	N/A	0
Ra4	N/A	N/A	N/A	0

LINKS

- The Virtual Foundry
- Bronze Filament Safety Data Sheet
- Stainless Steel Filament Safety Data Sheet
- Bronze Filament Technical Data Sheet
- Stainless Steel Filament Technical Data Sheet
- Filawarmer